## APPENDIX I: PAPUA NEW GUINEA: DEBT SUSTAINABILITY ANALYSIS

### I. INTRODUCTION

**Papua New Guinea has made significant progress in reducing its public debt burden.** However, Papua New Guinea still remains at moderate risk of external debt distress over the medium and long term. External debt burden indicators on public and publicly guaranteed (PPG) external debt are below their indicative thresholds under the baseline scenario, but some indicators would breach their thresholds if the terms of trade deteriorates further or LNG project related contingent liabilities are materialized.[1] Compared to the 2007 Article IV DSA, given the change in the global outlook, this analysis shows that Papua New Guinea exhibits more vulnerability to external shocks. The inclusion of domestic debt as well as several contingent liabilities also shows risks to long-run fiscal sustainability. Box 1 summarizes the macroeconomic framework underlying the baseline analysis.

### II. PUBLIC DEBT SUSTAINABILITY ANALYSIS[2]

**The public sector debt has been declining over the past 7 years** (Table 1) from about 71½ percent in 2002 to under 34 percent of GDP in 2007. This improvement reflects prudent fiscal policy, favorable terms of trade, appreciation of the kina against the U.S. dollar, lower interest rates, and sustained economic growth since 2003. The public external debt also declined from the peak of over 50 percent of GDP in 2001 to below 17 percent at end-2007.[3] External debt to multilateral institutions accounts for about 65 percent of public external debt, with Japan as the main bilateral creditor. Private sector external debt is estimated at about 14 percent of GDP at end-2007.

**The baseline scenario envisages a gradual increase in the public debt-to-GDP ratio in the medium term.**[4] It reflects the additional financing needs on the external side, mainly due to the sharp deterioration of the terms of trade and corresponding current account. However, all debt burden indicators remain well below the thresholds and are expected to decline in the long run. On the domestic side, the expectation of continued prudent fiscal policy and

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[1] Papua New Guinea is rated as a weak performer for its policies and institutions for the purposes of the IMF-WB low-income country DSA framework. Consequently, the applicable thresholds for this category to external public debt are: 30 percent for the present value (PV) of debt-to-GDP ratio; 100 percent for the PV of debt-to-exports ratio, 200 percent for the PV of debt-to-revenue ratio, 15 percent for the debt service-to-exports ratio, and 25 percent for the debt service-to-revenue ratio.

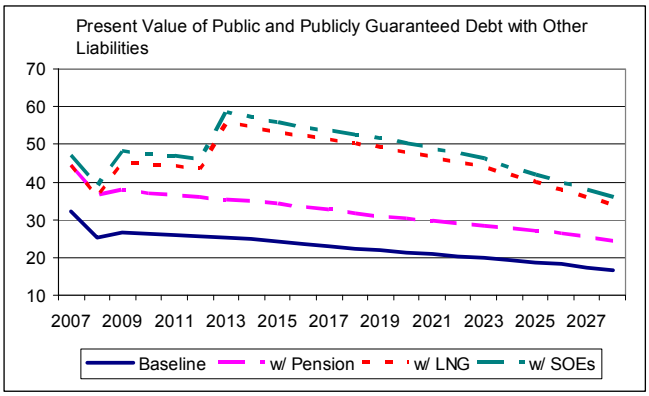
[2] Public debt includes domestic central government debt and external public and publicly guaranteed debt.

[3] By end-2008, it is estimated that public debt declined further to about 26½ percent of GDP with public external debt about 12 percent.

[4] Baseline scenario does not include the LNG project.

favorable debt dynamics are expected to help the public domestic debt burden continue to decline in the medium and long term. In sum, the public debt-to-GDP ratio is projected to peak at 30 percent in the medium term, then gradually decline to 22½ percent in the long run.

**The standard alternative scenarios and bound tests indicate that the projected debt path is particularly sensitive to changes in real GDP growth and further deterioration of terms of trade** (Table 2 and Figure 2). A shock to real GDP growth during 2009–10, equal to the historical average of GDP growth rate minus one standard deviation, would bring the present value (PV) of the public debt-to-GDP ratio to 71 percent in 2028. This highlights the need to manage expenditures prudently, while protecting priority spending, in the event of a growth slowdown. Further deterioration of the terms of trade would also bring the public debt burden to the alarming level, as discussed more detail below.




**Fiscal sustainability is sensitive to other liabilities arising from several sources.**[5] Unfunded civil service liabilities, amounting to be about 11 percent of GDP at end-2008, will remain sizeable though the government has made partial repayment recently. LNG project related contingent liabilities are estimated to be, in PV term, about 20½ percent of GDP in 2013.[6] Borrowings by SOEs are also estimated to be significant and growing, although information is limited.[7] Other contingent liabilities arising from the government's commitments under the memorandum of agreement with landowners are also reported to be significant and the government needs to pay some liabilities in near future to allow the LNG project to move forward.

[5] Only unfunded pension liabilities are included in Table 1 since other liabilities are all Fund staff estimates.

[6] These liabilities are arising from the issue of exchangeable bonds to finance the state's equity participation (in 2009) and the state's completion guarantee for debt financing (by 2013).Under this scheme, the creditor acquires exchangeable bonds, amounting to US$1.1 billion, with the option to exchange the bonds for equities in the Oil Search company. If the share price remains below this strike price, the state will have to pay to the creditor the difference between the market price and the face value of the bond. Under the completion guarantee, if the project fails, the government will have to pay up to US$2 billion to the creditor.

[7] Preliminary estimates for SOE related contingent liabilities, based on end-2007 financial statements of nine SOEs, are about 2.5 percent of GDP when we consider borrowings, various provisions, and other liabilities. However, there is no information on the off-balance sheet liabilities which also may be significant.

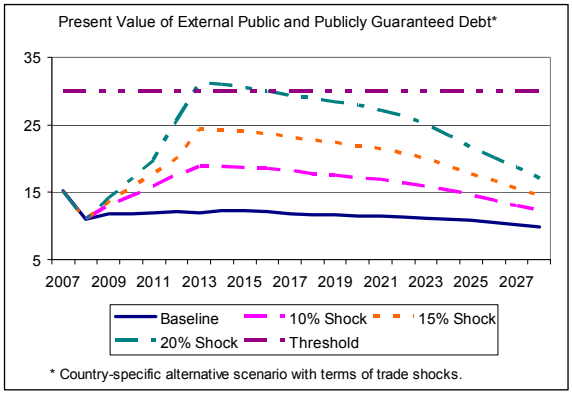
## III. External Debt Sustainability Analysis

**All PPG external debt and debt service indicators remain below the policy-dependent debt burden thresholds under the baseline scenario.** However, a country-specific scenario indicates the vulnerability of the external debt sustainability if the terms of trade deteriorates further in the medium term. The main results of the external DSA are as follows:

- **The debt burden indicators in the baseline scenario are expected to increase in the medium term before declining in the long run** (Table 3). Additional financing needs arising from the deterioration of the terms of trade and corresponding current account would raise the PV of PPG external debt over GDP from below 11 percent in 2008 to 12 percent in the medium term, which is still well below the 30 percent of threshold. The PVs of PPG external debt-to-exports and debt-to-revenue ratios are also expected to increase accordingly in the medium term, but still well below the indicative threshold levels.

- **The debt burden is expected to increase under the various standard stress tests, but all debt burden indicators remain below their indicative thresholds** (Table 4 and Figure 1). Alternative scenarios with key variables at their historical averages in 2008–28 show a reduction of the debt burden because of the current account surpluses and FDI inflows in recent commodity booming years.[8]

- **A country-specific scenario shows that the external debt sustainability would be vulnerable to the terms of trade shock.** If export prices decline further, by 20 percent compared to the January 2009 WEO price assumption for 2009–13, the PV of external PPG debt-to-GDP ratio is projected to breach the 30 percent threshold in the medium term, due to the deterioration of current account and corresponding additional




[8] Bound test B5, which combines four different shocks with smaller standard deviations than those shocks in tests B1–B4, also shows the similar patterns because recent rapid growth in exports during the commodity booming period generates rather improvement in the current account under this test.

financing needs.[9] The other debt and debt service indicators would rise significantly for the same period, though they are not expected to breach their indicative thresholds.

**Box 1. Macroeconomic Assumptions Underlying the DSA**

- **Real GDP growth** is projected to be at around 3.8 percent in the near term and then slow to about 2½ percent in the medium and long term—above the recent historical average of 2 percent. Moderate growth in the nonmineral sector, including in agriculture, and the several mining projects coming on stream will offset a steady decline in petroleum and copper production. However, the large Liquefied Natural Gas (LNG) project under consideration is not incorporated in the baseline scenario.
- **Inflation** is projected to peak in 2008 because of the rise in food and energy prices until July, and subsequently stabilize at around 2 percent in the long run.
- The **current account** surplus (including grants) will turn into a deficit in the near and medium term, primarily reflecting continued strong growth of capital imports related to investment projects as well as the impact of global economic slowdown.
- The **grant element of loans** is expected to decline. As per capita income rises, the share of external financing provided on concessional terms is expected to decline gradually over the projection period.[1]
- The **fiscal balance** is projected to turn into a deficit in 2009 and to remain in deficit over the medium term, due to falling mineral revenues and the spending of preceding years' mineral windfall revenues. The deficit is expected to be financed by a run down of resources held in trust accounts.

[1] Grant element of new public borrowing increases slightly in the near term, before it declines in the long run, due to the relatively large disbursement commitment from the AsDB in the pipeline.

The following text table summarizes Papua New Guinea's indicative thresholds, actual 2007 ratios, and average debt ratio dynamics in the medium and long term under the baseline scenario.

[9] Considering that PNG currently has comfortable level of reserves, we assume that additional financing needs would be filled by additional concessional external borrowings and the use of reserves. However, if the import coverage of reserves falls below 3 months, we assume that all financing needs should be filled by additional external borrowings.

Text Table: Policy-based PPG External Debt Burden Thresholds for PNG

| | Thresholds | PNG's ratios | | |
|---|---|---|---|---|
| | | 2007 | 2008–13 | 2014–28 |
| PV of debt in percent of | | | | |
| GDP | 30 | 15.2 | 11.8 | 11.3 |
| Exports | 100 | 19.2 | 19.3 | 22.6 |
| Revenues | 200 | 45.4 | 48.6 | 43.3 |
| Debt service in percent of | | | | |
| Exports | 15 | 4.1 | 2.1 | 1.6 |
| Revenues | 25 | 9.7 | 5.2 | 3.2 |

## IV. Conclusion

**Papua New Guinea still remains at moderate risk of external debt distress over the medium term.** Though external PPG debt burden indicators are well below their indicative thresholds under the baseline scenario, the projected PPG debt paths point to a significant vulnerability against the terms of trade shock and slower economic growth. The inclusion of the liabilities arising from several sources raises concerns about long-term fiscal sustainability.

**Continued prudent management of debt over the medium and long term will be required to mitigate the risks to public debt sustainability.** The staff encourages the authorities to build on recent steps and move forward to strengthen debt management capacity.

Table 1. Papua New Guinea: Public Sector Debt Sustainability Framework, Baseline Scenario, 2005–28

(In percent of GDP, unless otherwise indicated)

| | Actual | | | Average | Standard Deviation | Estimate | | Projections | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 2005 | 2006 | 2007 | | | 2008 | 2009 | 2010 | 2011 | 2012 | 2013 | 2008–13 Average | 2018 | 2028 | 2014–28 Average |
| Public sector debt 1/ | 47.7 | 39.1 | 33.7 | | | 26.7 | 29.5 | 29.4 | 29.6 | 29.9 | 29.8 | | 31.8 | 26.6 | |
| o/w foreign-currency denominated | 25.4 | 21.1 | 16.7 | | | 12.2 | 14.7 | 15.1 | 15.7 | 16.3 | 16.7 | | 21.1 | 19.9 | |
| Change in public sector debt | -12.3 | -8.6 | -5.4 | | | -7.0 | 2.8 | -0.1 | 0.2 | 0.3 | -0.1 | | 0.0 | -1.4 | |
| Identified debt-creating flows | -14.0 | -12.6 | -12.7 | | | -10.7 | 2.4 | 0.2 | 0.4 | -0.1 | 0.2 | | -0.8 | -0.6 | |
| Primary deficit | -5.9 | -8.5 | -10.2 | -3.9 | 3.6 | -6.3 | 0.4 | 0.2 | 0.0 | -0.6 | -0.4 | -1.1 | -0.5 | 0.3 | -0.4 |
| Revenue and grants | 34.9 | 36.7 | 37.3 | | | 32.7 | 27.1 | 26.5 | 26.6 | 27.0 | 27.3 | | 28.3 | 26.2 | |
| *of which: grants* | 8.4 | 5.3 | 3.9 | | | 3.5 | 3.3 | 3.3 | 3.4 | 3.5 | 3.6 | | 2.8 | 1.5 | |
| Primary (noninterest) expenditure | 29.0 | 28.2 | 27.2 | | | 26.4 | 27.5 | 26.7 | 26.5 | 26.4 | 26.9 | | 27.8 | 26.4 | |
| Automatic debt dynamics | -8.1 | -4.1 | -2.6 | | | -4.4 | 2.0 | 0.0 | 0.4 | 0.5 | 0.6 | | -0.3 | -0.9 | |
| Contribution from interest rate/growth differential | -3.9 | -1.6 | -1.1 | | | -2.1 | 0.9 | -0.4 | 0.0 | 0.1 | 0.4 | | -0.3 | -0.6 | |
| *of which: contribution from average real interest rate* | -1.8 | -0.4 | 1.3 | | | 0.1 | 1.9 | 0.7 | 0.9 | 1.0 | 1.1 | | 0.5 | 0.1 | |
| *of which: contribution from real GDP growth* | -2.1 | -1.2 | -2.4 | | | -2.2 | -1.0 | -1.1 | -1.0 | -0.9 | -0.7 | | -0.7 | -0.7 | |
| Contribution from real exchange rate depreciation | -4.2 | -2.5 | -1.5 | | | -2.3 | 1.2 | 0.4 | 0.4 | 0.3 | 0.2 | | ... | ... | |
| Other identified debt-creating flows | 0.0 | 0.0 | 0.0 | | | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | | 0.0 | 0.0 | |
| Privatization receipts (negative) | 0.0 | 0.0 | 0.0 | | | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | | 0.0 | 0.0 | |
| Recognition of implicit or contingent liabilities | 0.0 | 0.0 | 0.0 | | | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | | 0.0 | 0.0 | |
| Debt relief (HIPC and other) | 0.0 | 0.0 | 0.0 | | | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | | 0.0 | 0.0 | |
| Other (specify, e.g., bank recapitalization) | 0.0 | 0.0 | 0.0 | | | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | | 0.0 | 0.0 | |
| Residual, including asset changes | 1.7 | 4.0 | 7.3 | | | 3.7 | 0.4 | -0.3 | -0.2 | 0.4 | -0.3 | | 0.8 | -0.7 | |
| Other Sustainability Indicators | | | | | | | | | | | | | | | |
| PV of public sector debt | … | … | 32.2 | | | 25.4 | 26.6 | 26.1 | 25.8 | 25.7 | 25.2 | | 23.8 | 18.9 | |
| o/w foreign-currency denominated | ... | ... | 15.2 | | | 11.0 | 11.8 | 11.8 | 11.9 | 12.1 | 12.0 | | 13.1 | 12.2 | |
| o/w external | ... | ... | 15.2 | | | 11.0 | 11.8 | 11.8 | 11.9 | 12.1 | 12.0 | | 13.1 | 12.2 | |
| PV of unfunded pension liabilities (not included in public sector debt) | 10.7 | 9.5 | 12.3 | | | 11.0 | 11.1 | 10.7 | 10.4 | 10.1 | 9.9 | | 9.9 | 9.3 | |
| Gross financing need 2/ | 36.0 | 18.9 | 11.0 | | | 7.2 | 5.7 | 8.4 | 6.9 | 6.7 | 5.9 | | 3.8 | 2.6 | |
| PV of public sector debt-to-revenue and grants ratio (in percent) | … | … | 86.3 | | | 77.9 | 98.0 | 98.5 | 97.2 | 95.1 | 92.5 | | 83.9 | 72.2 | |
| PV of public sector debt-to-revenue ratio (in percent) | … | … | 96.2 | | | 87.3 | 111.7 | 112.7 | 111.6 | 109.2 | 106.3 | | 92.9 | 76.4 | |
| o/w external 3/ | … | … | 45.4 | | | 37.6 | 49.6 | 51.0 | 51.4 | 51.4 | 50.8 | | 51.3 | 49.3 | |
| Debt service-to-revenue and grants ratio (in percent) 4/ | 34.0 | 28.7 | 33.5 | | | 26.5 | 20.7 | 18.2 | 19.3 | 18.9 | 19.5 | | 12.8 | 11.2 | |
| Debt service-to-revenue ratio (in percent) 4/ | 12.2 | 8.7 | 9.6 | | | 9.3 | 9.0 | 7.6 | 8.4 | 8.4 | 9.0 | | 5.8 | 4.1 | |
| Primary deficit that stabilizes the debt-to-GDP ratio | 6.4 | 0.1 | -4.8 | | | 0.7 | -2.5 | 0.3 | -0.2 | -0.9 | -0.3 | | -0.5 | 1.6 | |
| Key macroeconomic and fiscal assumptions | | | | | | | | | | | | | | | |
| Real GDP growth (in percent) | 3.6 | 2.6 | 6.5 | 2.1 | 2.6 | 7.0 | 3.9 | 3.7 | 3.3 | 3.0 | 2.4 | 3.9 | 2.4 | 2.4 | 2.4 |
| Average nominal interest rate on forex debt (in percent) | 2.8 | 3.1 | 3.2 | 3.5 | 0.5 | 3.8 | 2.3 | 2.2 | 2.1 | 2.0 | 1.9 | 2.4 | 1.6 | 1.8 | 1.7 |
| Average real interest rate on domestic debt (in percent) | -7.9 | -4.0 | 5.7 | 2.6 | 6.8 | -2.0 | 12.9 | 4.6 | 6.7 | 7.5 | 8.6 | 6.4 | 4.9 | 2.0 | 4.3 |
| Real exchange rate depreciation (in percent, + indicates depreciation) | -12.5 | -9.9 | -7.4 | -4.6 | 11.1 | -14.4 | ... | ... | ... | ... | ... | ... | ... | ... | ... |
| Inflation rate (GDP deflator, in percent) | 15.9 | 9.9 | 2.6 | 8.1 | 5.4 | 11.1 | -3.3 | 2.5 | 2.0 | 2.0 | 2.4 | 2.8 | 3.7 | 5.6 | 4.0 |
| Growth of real primary spending (deflated by GDP deflator, in percent) | 2.3 | -0.4 | 2.7 | 1.5 | 10.5 | 3.8 | 8.3 | 0.9 | 2.6 | 2.5 | 4.2 | 3.7 | 2.6 | 1.5 | 2.3 |
| Grant element of new external borrowing (in percent) | ... | ... | ... | … | … | 43.5 | 43.3 | 46.3 | 45.6 | 46.2 | 46.3 | 45.2 | 37.5 | 25.4 | ... |

Sources: Country authorities; and Fund staff estimates and projections.

1/ Public debt includes gross debt of central government, central bank, and statutory authorities.

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 2. Papua New Guinea: Sensitivity Analysis for Key Indicators of Public Debt 2008–28

| | | Projections | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 2008 | 2009 | 2010 | 2011 | 2012 | 2013 | 2018 | 2028 |
| **PV of Debt-to-GDP Ratio** | | | | | | | | |
| **Baseline** | 25 | 27 | 26 | 26 | 26 | 25 | 24 | 19 |
| **A. Alternative scenarios** | | | | | | | | |
| A1. Real GDP growth and primary balance are at historical averages | 25 | 23 | 18 | 14 | 11 | … | … | … |
| A2. Primary balance is unchanged from 2008 | 25 | 20 | 13 | … | … | … | … | … |
| A3. Permanently lower GDP growth 1/ | 25 | 27 | 27 | 27 | 28 | 28 | 33 | 57 |
| **B. Bound tests** | | | | | | | | |
| B1. Real GDP growth is at historical average minus one standard deviations in 2009–10 | 25 | 29 | 32 | 33 | 35 | 37 | 48 | 73 |
| B2. Primary balance is at historical average minus one standard deviations in 2009–10 | 25 | 26 | 25 | 25 | 24 | 24 | 22 | 17 |
| B3. Combination of B1-B2 using one half standard deviation shocks | 25 | 25 | 23 | 24 | 25 | 26 | 32 | 46 |
| B4. One-time 30 percent real depreciation in 2009 | 25 | 31 | 31 | 30 | 30 | 29 | 27 | 28 |
| B5. 10 percent of GDP increase in other debt-creating flows in 2009 | 25 | 37 | 36 | 36 | 36 | 36 | 36 | 33 |
| **PV of Debt-to-Revenue Ratio 2/** | | | | | | | | |
| **Baseline** | 78 | 98 | 98 | 97 | 95 | 92 | 84 | 72 |
| **A. Alternative scenarios** | | | | | | | | |
| A1. Real GDP growth and primary balance are at historical averages | 78 | 84 | 69 | 54 | 40 | … | … | … |
| A2. Primary balance is unchanged from 2008 | 78 | 74 | 49 | … | … | … | … | … |
| A3. Permanently lower GDP growth 1/ | 78 | 99 | 101 | 102 | 102 | 102 | 117 | 217 |
| **B. Bound tests** | | | | | | | | |
| B1. Real GDP growth is at historical average minus one standard deviations in 2009–10 | 78 | 106 | 118 | 124 | 129 | 134 | 169 | 277 |
| B2. Primary balance is at historical average minus one standard deviations in 2009–10 | 78 | 95 | 94 | 92 | 90 | 88 | 79 | 65 |
| B3. Combination of B1-B2 using one half standard deviation shocks | 78 | 91 | 85 | 89 | 92 | 94 | 113 | 173 |
| B4. One-time 30 percent real depreciation in 2009 | 78 | 115 | 116 | 114 | 111 | 108 | 97 | 108 |
| B5. 10 percent of GDP increase in other debt-creating flows in 2009 | 78 | 135 | 136 | 136 | 134 | 131 | 126 | 128 |
| **Debt Service-to-Revenue Ratio 2/** | | | | | | | | |
| **Baseline** | 27 | 21 | 18 | 19 | 19 | 20 | 13 | 11 |
| **A. Alternative scenarios** | | | | | | | | |
| A1. Real GDP growth and primary balance are at historical averages | 27 | 21 | 18 | 6 | 3 | … | … | … |
| A2. Primary balance is unchanged from 2008 | 27 | 21 | 16 | … | … | … | … | … |
| A3. Permanently lower GDP growth 1/ | 27 | 21 | 18 | 20 | 20 | 22 | 24 | 70 |
| **B. Bound tests** | | | | | | | | |
| B1. Real GDP growth is at historical average minus one standard deviations in 2009–10 | 27 | 21 | 20 | 24 | 28 | 32 | 45 | 100 |
| B2. Primary balance is at historical average minus one standard deviations in 2009–10 | 27 | 21 | 18 | 17 | 17 | 17 | 10 | 8 |
| B3. Combination of B1-B2 using one half standard deviation shocks | 27 | 21 | 18 | 12 | 11 | 16 | 21 | 54 |
| B4. One-time 30 percent real depreciation in 2009 | 27 | 22 | 20 | 22 | 23 | 24 | 22 | 36 |
| B5. 10 percent of GDP increase in other debt-creating flows in 2009 | 27 | 21 | 21 | 49 | 28 | 44 | 31 | 37 |

Sources: Country authorities; and Fund staff estimates and projections.

1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the length of the projection period.

2/ Revenues are defined inclusive of grants.

Table 3. Papua New Guinea: External Debt Sustainability Framework, Baseline Scenario, 2005–28 1/

(In percent of GDP, unless otherwise indicated)

| | Actual | | | Historical Average | Standard Deviation | Projections | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 2005 | 2006 | 2007 | | | 2008 | 2009 | 2010 | 2011 | 2012 | 2013 | 2008–13 Average | 2018 | 2028 | 2014–28 Average |
| **External debt (nominal) 1/** | **41.2** | **38.6** | **30.7** | | | **24.4** | **27.9** | **29.5** | **31.2** | **33.2** | **34.7** | | **37.8** | **35.0** | |
| o/w public and publicly guaranteed (PPG) | 25.4 | 21.1 | 16.7 | | | 12.2 | 14.7 | 15.1 | 15.7 | 16.3 | 16.7 | | 21.1 | 19.9 | |
| Change in external debt | -10.6 | -2.6 | -7.9 | | | -6.3 | 3.5 | 1.5 | 1.7 | 2.0 | 1.5 | | 0.3 | -1.1 | |
| Identified net debt-creating flows | -14.2 | -10.4 | -12.2 | | | -7.9 | 2.5 | 0.4 | 0.0 | -0.5 | -1.2 | | -3.1 | -1.4 | |
| **Non-interest current account deficit** | **-5.5** | **-4.9** | **-4.0** | **-6.0** | **2.9** | **-4.7** | **4.9** | **2.8** | **2.3** | **1.8** | **0.8** | | **-2.4** | **-1.5** | -1.8 |
| Deficit in balance of goods and services | -2.9 | -9.3 | -5.7 | | | -6.0 | 4.8 | 2.8 | 2.5 | 2.1 | 1.3 | | -11.9 | -8.5 | |
| Exports | 71.7 | 79.9 | 79.1 | | | 70.6 | 54.9 | 58.9 | 60.7 | 61.5 | 61.6 | | 53.4 | 38.8 | |
| Imports | 68.8 | 70.6 | 73.5 | | | 64.5 | 59.7 | 61.8 | 63.2 | 63.5 | 62.9 | | 41.5 | 30.3 | |
| Net current transfers (negative = inflow) | -8.9 | -4.9 | -5.5 | -5.8 | 1.3 | -3.9 | -3.7 | -3.8 | -3.9 | -4.0 | -4.1 | | -3.2 | -1.8 | -2.8 |
| o/w official | -8.0 | -5.3 | -4.9 | | | -3.5 | -3.3 | -3.3 | -3.4 | -3.5 | -3.6 | | -2.8 | -1.6 | |
| Other current account flows (negative = net inflow) | 6.3 | 9.3 | 7.1 | | | 5.3 | 3.8 | 3.8 | 3.7 | 3.7 | 3.6 | | 12.7 | 8.7 | |
| **Net FDI (negative = inflow)** | **-1.4** | **-3.4** | **-7.2** | **-3.4** | **2.7** | **-3.4** | **-3.2** | **-3.3** | **-3.4** | **-3.5** | **-3.6** | | **-2.1** | **-1.2** | -1.9 |
| **Endogenous debt dynamics 2/** | **-7.4** | **-2.0** | **-0.9** | | | **0.1** | **0.8** | **0.9** | **1.0** | **1.2** | **1.5** | | **1.4** | **1.4** | |
| Contribution from nominal interest rate | 1.3 | 2.6 | 2.3 | | | 1.9 | 1.7 | 1.8 | 2.0 | 2.1 | 2.3 | | 2.2 | 2.2 | |
| Contribution from real GDP growth | -1.6 | -0.9 | -2.3 | | | -1.8 | -1.0 | -1.0 | -0.9 | -0.9 | -0.8 | | -0.8 | -0.8 | |
| Contribution from price and exchange rate changes | -7.1 | -3.7 | -1.0 | | | … | … | … | … | … | … | | … | … | |
| **Residual (3–4) 3/** | **3.6** | **7.8** | **4.2** | | | **1.6** | **1.0** | **1.2** | **1.8** | **2.4** | **2.8** | | **3.4** | **0.3** | |
| o/w exceptional financing | 0.0 | 0.0 | 0.0 | | | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | | 0.0 | 0.0 | |
| PV of external debt 4/ | … | … | 29.2 | | | 23.1 | 25.0 | 26.2 | 27.4 | 28.9 | 30.1 | | 29.8 | 27.3 | |
| In percent of exports | … | … | 36.9 | | | 32.8 | 45.5 | 44.4 | 45.2 | 47.0 | 48.8 | | 55.7 | 70.4 | |
| **PV of PPG external debt** | … | … | **15.2** | | | **11.0** | **11.8** | **11.8** | **11.9** | **12.1** | **12.0** | | **13.1** | **12.2** | |
| **In percent of exports** | … | … | **19.2** | | | **15.5** | **21.5** | **20.1** | **19.6** | **19.7** | **19.5** | | **24.5** | **31.4** | |
| **In percent of government revenues** | … | … | **45.4** | | | **37.6** | **49.6** | **51.0** | **51.4** | **51.4** | **50.8** | | **51.3** | **49.3** | |
| **Debt service-to-exports ratio (in percent)** | **9.7** | **8.3** | **8.4** | | | **6.4** | **7.4** | **7.3** | **7.3** | **7.3** | **8.7** | | **9.7** | **13.3** | |
| **PPG debt service-to-exports ratio (in percent)** | **6.0** | **3.5** | **4.1** | | | **2.5** | **2.1** | **2.1** | **2.0** | **1.9** | **1.9** | | **1.7** | **2.6** | |
| **PPG debt service-to-revenue ratio (in percent)** | **16.4** | **9.0** | **9.7** | | | **6.0** | **4.9** | **5.3** | **5.3** | **5.0** | **5.0** | | **3.5** | **4.0** | |
| Total gross financing need (Billions of U.S. dollars) | 0.0 | -0.1 | -0.3 | | | -0.3 | 0.5 | 0.3 | 0.3 | 0.2 | 0.2 | | 0.1 | 0.5 | |
| Non-interest current account deficit that stabilizes debt ratio | 5.2 | -2.3 | 3.9 | | | 1.6 | 1.4 | 1.3 | 0.6 | -0.2 | -0.7 | | -2.7 | -0.4 | |
| **Key macroeconomic assumptions** | | | | | | | | | | | | | | | |
| Real GDP growth (in percent) | 3.6 | 2.6 | 6.5 | 1.4 | 3.5 | 7.0 | 3.9 | 3.7 | 3.3 | 3.0 | 2.4 | 3.9 | 2.4 | 2.4 | 2.4 |
| GDP deflator in U.S. dollar terms (change in percent) | 15.9 | 9.9 | 2.6 | 8.4 | 5.2 | 11.1 | -3.3 | 2.5 | 2.0 | 2.0 | 2.4 | 2.8 | 3.7 | 5.6 | 4.0 |
| Effective interest rate (percent) 5/ | 3.0 | 7.2 | 6.6 | 4.5 | 1.4 | 7.5 | 7.1 | 7.0 | 7.0 | 7.1 | 7.1 | 7.1 | 6.3 | 6.5 | 6.4 |
| Growth of exports of G&S (U.S. dollar terms, in percent) | 27.3 | 27.6 | 12.9 | 6.5 | 18.7 | 13.0 | -21.1 | 9.8 | 5.5 | 3.8 | 3.2 | 2.4 | 3.6 | 2.9 | 3.0 |
| Growth of imports of G&S (U.S. dollar terms, in percent) | 35.6 | 17.5 | 18.5 | 9.4 | 20.5 | 11.3 | -6.2 | 5.8 | 4.8 | 3.1 | 1.9 | 3.5 | 2.9 | 3.0 | 1.3 |
| Grant element of new public sector borrowing (in percent) | … | … | … | … | … | 43.5 | 43.3 | 46.3 | 45.6 | 46.2 | 46.3 | 45.2 | 37.5 | 25.4 | 33.8 |
| Government revenues (excluding grants, in percent of GDP) | 26.5 | 31.4 | 33.5 | | | 29.1 | 23.8 | 23.2 | 23.2 | 23.5 | 23.7 | | 25.6 | 24.7 | 25.5 |
| Aid flows (in Billions of U.S. dollars) 7/ | 0.5 | 0.3 | 0.3 | | | 0.3 | 0.5 | 0.4 | 0.4 | 0.5 | 0.5 | | 0.5 | 0.4 | |
| o/w Grants | 0.4 | 0.3 | 0.2 | | | 0.3 | 0.3 | 0.3 | 0.3 | 0.3 | 0.3 | | 0.3 | 0.3 | |
| o/w Concessional loans | 0.0 | 0.0 | 0.0 | | | 0.0 | 0.2 | 0.1 | 0.2 | 0.2 | 0.1 | | 0.2 | 0.1 | |
| Grant-equivalent financing (in percent of GDP) 8/ | … | … | … | | | 3.6 | 4.5 | 4.2 | 4.3 | 4.4 | 4.3 | | 3.4 | 1.7 | 2.9 |
| Grant-equivalent financing (in percent of external financing) 8/ | … | … | … | | | 96.7 | 73.8 | 81.6 | 81.0 | 80.4 | 82.6 | | 75.3 | 71.2 | 73.8 |
| *Memorandum items:* | | | | | | | | | | | | | | | |
| Nominal GDP (Billions of U.S. dollars) | 4.9 | 5.6 | 6.4 | | | 8.1 | 8.2 | 8.4 | 8.6 | 8.8 | 9.1 | | 12.3 | 22.5 | |
| Nominal dollar GDP growth | 24.7 | 14.4 | 13.9 | | | 26.7 | 1.3 | 2.3 | 2.4 | 2.6 | 2.9 | 6.4 | 6.2 | 6.2 | 6.2 |

Source: Staff simulations.

1/ Includes both public and private sector external debt.
2/ Derived as [i - g - r(1+g)]/(1+g+r+gr) times previous period debt ratio, with i = nominal interest rate; g = real GDP growth rate, and r = growth rate of GDP deflator in U.S. dollar terms.
3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.
4/ Assumes that PV of private sector debt is equivalent to its face value.
5/ Current-year interest payments divided by previous period debt stock.
6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.
7/ Defined as grants, concessional loans, and debt relief.
8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 4a. Papua New Guinea: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2008–28
(In percent)

| | | | | Projections | | | | |
|---|---|---|---|---|---|---|---|---|
| | 2008 | 2009 | 2010 | 2011 | 2012 | 2013 | **2018** | 2028 |
| **PV of debt-to GDP ratio** | | | | | | | | |
| **Baseline** | 11 | 12 | 12 | 12 | 12 | 12 | **13** | 12 |
| **A. Alternative Scenarios** | | | | | | | | |
| A1. Key variables at their historical averages in 2008–28 1/ | 11 | 4 | … | … | … | … | … | … |
| A2. New public sector loans on less favorable terms in 2008–28 2/ | 11 | 12 | 12 | 11 | 11 | 11 | **11** | 8 |
| A3. Deterioration of the terms of trade in 2009-13 3/ | 11 | 14 | 17 | 20 | 26 | 31 | **29** | 19 |
| **B. Bound Tests** | | | | | | | | |
| B1. Real GDP growth at historical average minus one standard deviation in 2009–10 | 11 | 12 | 13 | 13 | 14 | 14 | **15** | 12 |
| B2. Export value growth at historical average minus one standard deviation in 2009–10 4/ | 11 | 6 | 10 | 11 | 11 | 11 | **12** | 11 |
| B3. U.S. dollar GDP deflator at historical average minus one standard deviation in 2009–10 | 11 | 11 | 11 | 12 | 12 | 12 | **13** | 10 |
| B4. Net non-debt creating flows at historical average minus one standard deviation in 2009–10 5/ | 11 | 13 | 14 | 15 | 15 | 15 | **15** | 11 |
| B5. Combination of B1-B4 using one-half standard deviation shocks | 11 | … | … | … | … | … | … | … |
| B6. One-time 30 percent nominal depreciation relative to the baseline in 2009 6/ | 11 | 17 | 17 | 18 | 18 | 18 | **19** | 16 |
| **PV of debt-to-exports ratio** | | | | | | | | |
| **Baseline** | 16 | 22 | 20 | 20 | 20 | 20 | **25** | 31 |
| **A. Alternative Scenarios** | | | | | | | | |
| A1. Key variables at their historical averages in 2008–28 1/ | 16 | 7 | … | … | … | … | … | … |
| A2. New public sector loans on less favorable terms in 2008–28 2/ | 16 | 22 | 20 | 19 | 18 | 18 | **20** | 20 |
| A3. Deterioration of the terms of trade in 2009-13 3/ | 16 | 32 | 36 | 41 | 53 | 64 | **56** | 47 |
| **B. Bound Tests** | | | | | | | | |
| B1. Real GDP growth at historical average minus one standard deviation in 2009–10 | 16 | 21 | 21 | 21 | 22 | 23 | **28** | 32 |
| B2. Export value growth at historical average minus one standard deviation in 2009–10 4/ | 16 | 10 | 19 | 20 | 21 | 21 | **27** | 33 |
| B3. U.S. dollar GDP deflator at historical average minus one standard deviation in 2009–10 | 16 | 21 | 21 | 21 | 22 | 23 | **28** | 32 |
| B4. Net non-debt creating flows at historical average minus one standard deviation in 2009–10 5/ | 16 | 23 | 25 | 25 | 26 | 27 | **32** | 34 |
| B5. Combination of B1-B4 using one-half standard deviation shocks | 16 | … | … | … | … | … | … | … |
| B6. One-time 30 percent nominal depreciation relative to the baseline in 2009 6/ | 16 | 21 | 21 | 21 | 22 | 23 | **28** | 32 |
| **PV of debt-to-revenue ratio** | | | | | | | | |
| **Baseline** | 38 | 50 | 51 | 51 | 51 | 51 | **51** | 49 |
| **A. Alternative Scenarios** | | | | | | | | |
| A1. Key variables at their historical averages in 2008–28 1/ | 38 | 16 | … | … | … | … | … | … |
| A2. New public sector loans on less favorable terms in 2008–28 2/ | 38 | 50 | 50 | 49 | 48 | 46 | **42** | 32 |
| A3. Deterioration of the terms of trade in 2009-13 3/ | 35 | 49 | 62 | 75 | 99 | 120 | **110** | 70 |
| **B. Bound Tests** | | | | | | | | |
| B1. Real GDP growth at historical average minus one standard deviation in 2009–10 | 38 | 52 | 56 | 58 | 58 | 58 | **57** | 48 |
| B2. Export value growth at historical average minus one standard deviation in 2009–10 4/ | 38 | 27 | 44 | 46 | 47 | 47 | **49** | 43 |
| B3. U.S. dollar GDP deflator at historical average minus one standard deviation in 2009–10 | 38 | 47 | 49 | 50 | 51 | 51 | **49** | 41 |
| B4. Net non-debt creating flows at historical average minus one standard deviation in 2009–10 5/ | 38 | 54 | 62 | 63 | 63 | 63 | **60** | 46 |
| B5. Combination of B1-B4 using one-half standard deviation shocks | 38 | … | … | … | … | … | … | … |
| B6. One-time 30 percent nominal depreciation relative to the baseline in 2009 6/ | 38 | 72 | 75 | 77 | 78 | 78 | **76** | 64 |

Table 4b. Papua New Guinea: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2008–28 (continued)

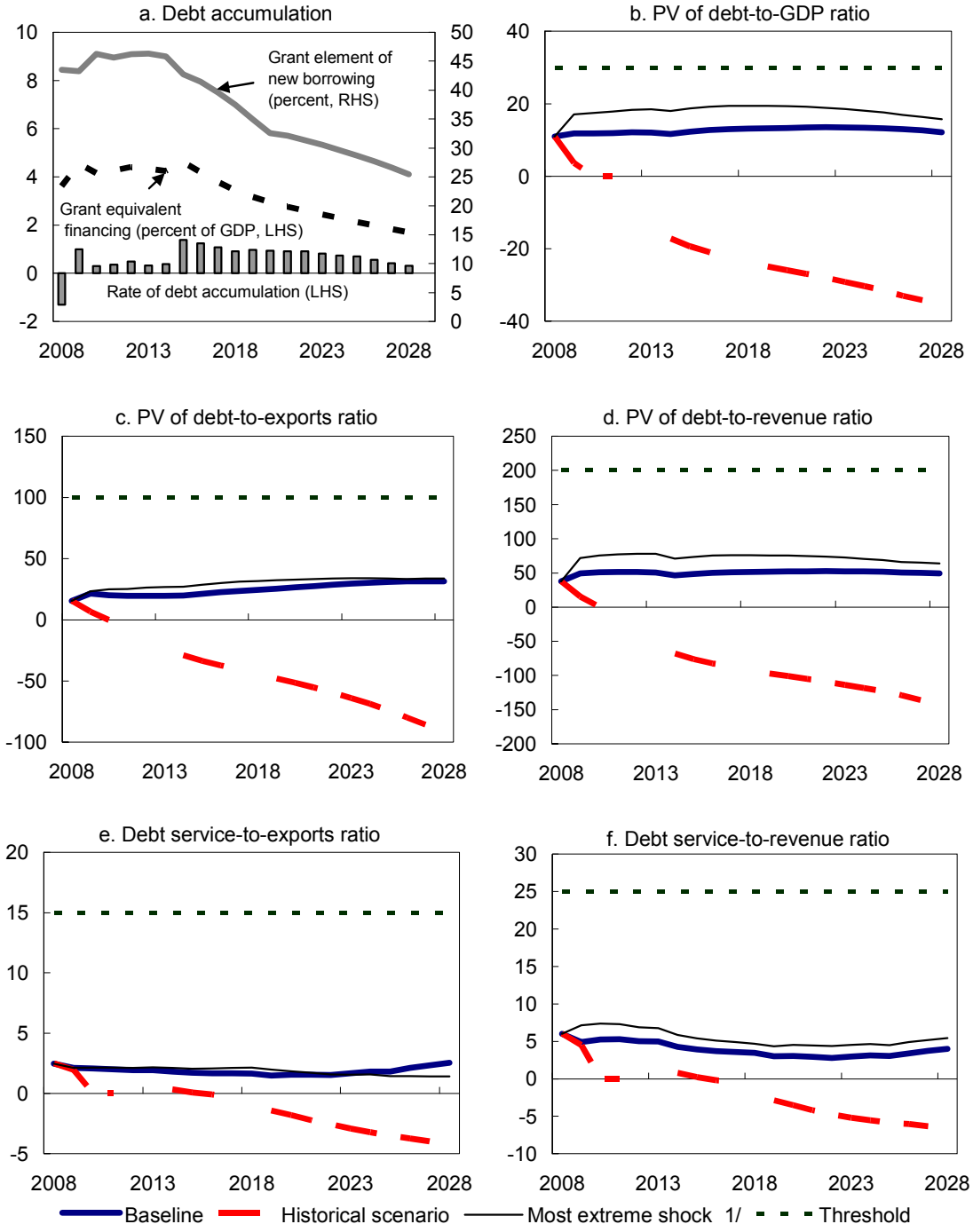
(In percent)

| | Projections | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 2008 | 2009 | 2010 | 2011 | 2012 | 2013 | **2018** | 2028 |
| **Debt service-to-exports ratio** | | | | | | | | |
| **Baseline** | 2 | 2 | 2 | 2 | 2 | 2 | **2** | 3 |
| **A. Alternative Scenarios** | | | | | | | | |
| A1. Key variables at their historical averages in 2008–28 1/ | 2 | 2 | … | … | … | … | … | … |
| A2. New public sector loans on less favorable terms in 2008–28 2/ | 2 | 2 | 2 | 2 | 2 | 2 | **2** | 1 |
| A3. Deterioration of the terms of trade in 2009-13 3/ | 2 | 3 | 3 | 3 | 3 | 3 | **2** | 5 |
| **B. Bound Tests** | | | | | | | | |
| B1. Real GDP growth at historical average minus one standard deviation in 2009–10 | 2 | 2 | 2 | 2 | 2 | 2 | **2** | 3 |
| B2. Export value growth at historical average minus one standard deviation in 2009–10 4/ | 2 | 2 | 2 | 2 | 2 | 2 | **1** | 3 |
| B3. U.S. dollar GDP deflator at historical average minus one standard deviation in 2009–10 | 2 | 2 | 2 | 2 | 2 | 2 | **2** | 3 |
| B4. Net non-debt creating flows at historical average minus one standard deviation in 2009–10 5/ | 2 | 2 | 2 | 2 | 2 | 2 | **2** | 3 |
| B5. Combination of B1-B4 using one-half standard deviation shocks | 2 | … | … | … | … | … | … | … |
| B6. One-time 30 percent nominal depreciation relative to the baseline in 2009 6/ | 2 | 2 | 2 | 2 | 2 | 2 | **2** | 3 |
| **Debt service-to-revenue ratio** | | | | | | | | |
| **Baseline** | 6 | 5 | 5 | 5 | 5 | 5 | **3** | 4 |
| **A. Alternative Scenarios** | | | | | | | | |
| A1. Key variables at their historical averages in 2008–28 1/ | 6 | 4 | … | … | … | … | … | … |
| A2. New public sector loans on less favorable terms in 2008–28 2/ | 6 | 5 | 5 | 5 | 5 | 5 | **4** | 2 |
| A3. Deterioration of the terms of trade in 2009-13 3/ | 6 | 4 | 5 | 5 | 5 | 6 | **5** | 8 |
| **B. Bound Tests** | | | | | | | | |
| B1. Real GDP growth at historical average minus one standard deviation in 2009–10 | 6 | 5 | 6 | 5 | 5 | 5 | **4** | 4 |
| B2. Export value growth at historical average minus one standard deviation in 2009–10 4/ | 6 | 5 | 5 | 5 | 5 | 5 | **2** | 4 |
| B3. U.S. dollar GDP deflator at historical average minus one standard deviation in 2009–10 | 6 | 5 | 5 | 5 | 4 | 4 | **3** | 4 |
| B4. Net non-debt creating flows at historical average minus one standard deviation in 2009–10 5/ | 6 | 5 | 5 | 5 | 5 | 5 | **4** | 4 |
| B5. Combination of B1-B4 using one-half standard deviation shocks | 6 | … | … | … | … | … | … | … |
| B6. One-time 30 percent nominal depreciation relative to the baseline in 2009 6/ | 6 | 7 | 7 | 7 | 7 | 7 | **5** | 5 |
| Memorandum item: | | | | | | | | |
| Grant element assumed on residual financing (i.e., financing required above baseline) 7/ | 35 | 35 | 35 | 35 | 35 | 35 | **35** | 35 |

Source: Staff projections and simulations.

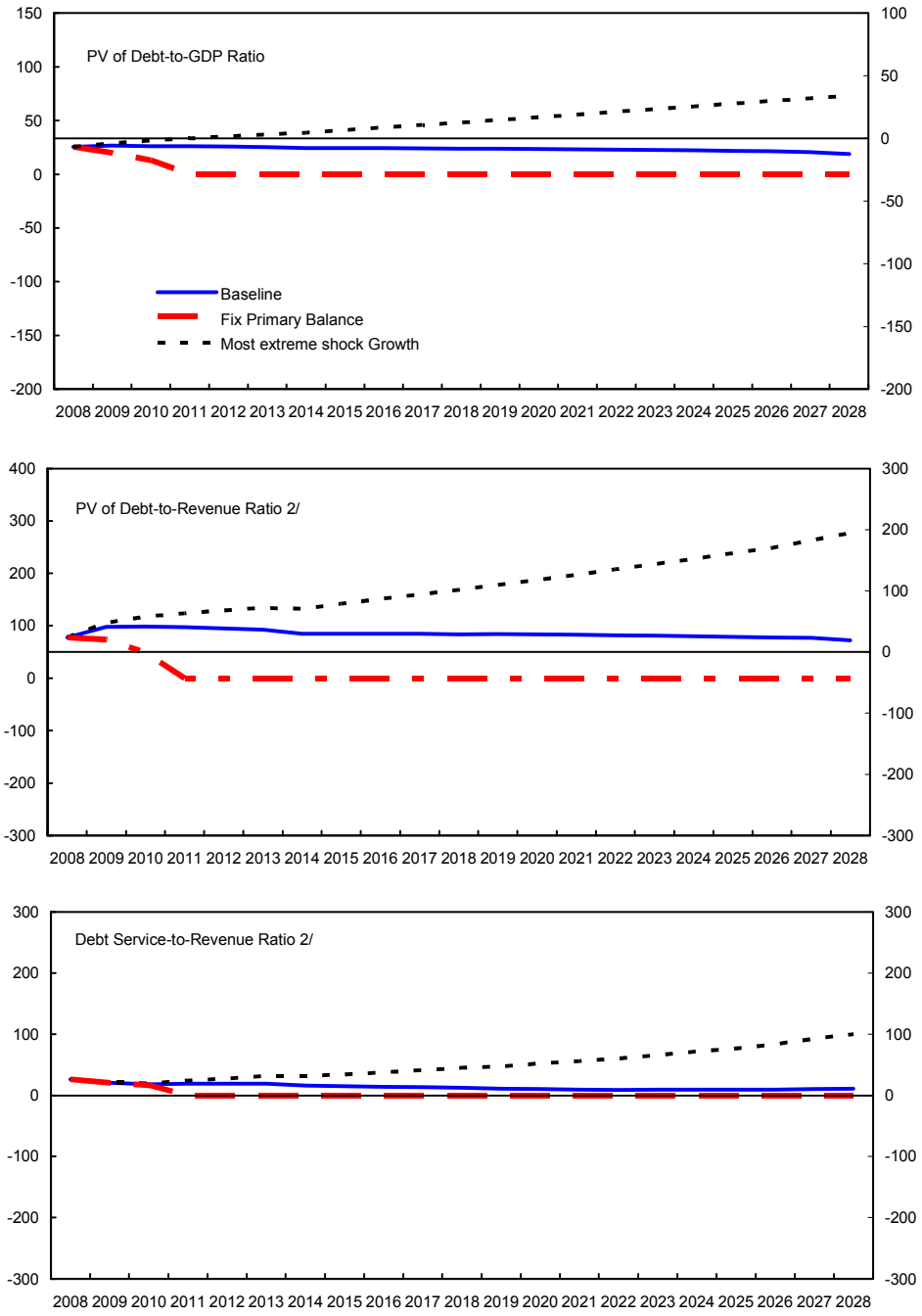
1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Export prices are assumed to be lower than the baseline by 20% in 2009-13.
4/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
5/ Includes official and private transfers and FDI.
6/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
7/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Figure 1. Papua New Guinea: Indicators of Public and Publicly Guaranteed External Debt under Standard Alternatives Scenarios, 2008–28 1/




Source: Staff projections and simulations.
1/ The most extreme stress test is the test that yields the highest ratio in 2018.

Figure 2. Papua New Guinea: Indicators of Public Debt Under Standard Alternative Scenarios, 2008–28 1/




Sources: Country authorities; and Fund staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio in 2018.
2/ Revenues are defined inclusive of grants.